Iteration 4 JAMES

$\hat{y}_1 = (1.268)(1) + 1.914 = 3.182$

$\hat{y}_2 = (1.268)(3) + 1.914 = 5.718$

ERRORS

$e_1 = 3 - 3.182 = -0.182$

$e_2 = 6 - 5.718 = 0.282$

$MSE = 1/2[(-0.182)^2 + (0.282)^2] = 1/2[0.0331 + 0.0795] = MSE = 0.0563$

$dJ/dm = 10m + 4b - 21 = 10(1.268) + 4(1.914) - 21 = 12.68 + 7.656 - 21 = -0.664$

$dJ/db = 4m + 2b - 9 = 4(1.268) + 2(1.914) - 9 = 5.072 + 3.828 - 9 = -0.10$

$m_4 = 1.268 - (0.1)(-0.664) = 1.268 + 0.0664$

$= m_4 = 1.334$

$b_4 = 1.914 - (0.1)(-0.10) = 1.914 + 0.01 = b_4$

1.924

(Judith)

Second Iteration ($m_1 = 1.7$, $b_1 = 2.7$)

i) Calculating Predicted $\hat{y}_{n_2}$ " From $\hat{y}_n = m x_n + b$

Point remain fixed $(1, 3)$; $(3, 6)$
$x_1\ y_1$ $\quad$ $x_2\ y_2$

$\Rightarrow \hat{y}_{n_1} = m_1 x_1 + b_1$

$= (1.7)(1) + 2.7)$

$\boxed{\hat{y}_{n_1} = 3.8}$

$\Rightarrow \hat{y}_{n_2} = m_2 x_2 + b_1$

$= (1.7)(3) + (2.7)$

$\boxed{\hat{y}_{n_2} = 7.2}$

Finding errors, Error $(e_1) = y_1 - \hat{y}_2$

$\Rightarrow e_2 = (3 - 3.8)$, $e_2 = (6 - 7.2)$

$e_{2_n} = -0.8$ $\qquad$ $e_{2_n} = -1.8$

Finding mean squared Error,

$MSE = \frac{1}{n}[e_{i_n}^2 + e_{i_n+1} \ldots]$, For $n = 2$

$MSE = \frac{1}{2}[(-0.8)^2 + (-1.8)^2] = 3.88\ (\frac{1}{2})$

$MSE = 1.94$

Finding new m and new b,

at $m_1 = 1.7$, b_1 2.7

$m_2 = m_1 - \alpha \frac{dJ}{dm}$, but $\frac{dJ}{dm} = 10mt4b - 27$

$m_2 = (1.7) - \frac{4.4}{10} = 10(1.7) + 4(2.7) - 27$

$= 4.4$

Continuation $m_2 = (1.7) - \frac{1}{10}(4.4) = \boxed{m_1 = 1.26}$

new b = old b $- \alpha \frac{dJ}{db}$

but $\frac{dJ}{db} = 4m + 2b - 9$, $m_1 = 1.7$, $b_1 = 2.7$

$\frac{dJ}{db} = 4(1.7) + 2(2.7) - 9 = \boxed{\frac{dJ}{db} = 2.0}$

And So, $b_2 - b_1 - \alpha\ dJ/db = b_2 = 2.7 - \frac{1}{20}(2.0) = \boxed{b_2 = 1.9}$

Points x	y	m_0	b_0	m_1	b_1	b_2	dJ/dm	dJ/db	MSE
1	3	1	1	1.7	2.1	1.9	4.4	2.0	1.94
3	6								

Gradient Descent Formula (1) Group 5

model $\rightarrow y = mx + b$, $\alpha = 0.1$, $m = -1$, $b = 1$

$\cdot J(m,b) = \frac{1}{n}(e_i^2 + e_{i+1}^2 + \dots)$, Error $= y_i - \hat{y}$

Points: (1,3) & (3,6)

$e_1 = y_i - \hat{y}_i = y - (m+b)$
$= 3 - (m+b)$

$e_2 = y_i - \hat{y}_i = 6 - (3m+b)$

* Finding MSE

$\Rightarrow \frac{1}{n}[e_1^2 + e_2^2] = \frac{1}{2}[(3-(m+b))^2 + (6-(3m+b))^2]$

* Cost function

$J(m,b) = \frac{1}{2}[3-(m+b)]^2 + [6-(3m+b)]^2$
$= \frac{1}{2}[(3-m-b)^2 + (6-3m-b)^2]$ &

for $m = -1$, $b = 1$

$\rightarrow y(1,3)$: $\hat{y} = (-1)(1) + (1) = 0$ | $e_1 = 0 - 3 = \underline{-3}$

$\rightarrow y(3,6)$: $\hat{y} = (-1)(3) + (1) = -2$ | $e_2 = -2 - (-6) = \underline{-8}$

$\frac{\partial J}{\partial m} = \frac{2}{2}[(1)(-3) + (3)(-8)]$
$= -3 - 24$
$= \underline{-27}$

$\frac{\partial J}{\partial b} = \frac{2}{2}[-3 + (-8)]$
$= \underline{-11}$

$m = m - \alpha \frac{\partial J}{\partial m}$
$= -1 - (0.1)(-27)$
$\boxed{m_{new} = \underline{1.7}}$

$b = b - \alpha \frac{\partial J}{\partial b}$
$= 1 - (0.1)(-11)$
$\boxed{b_{new} = \underline{2.1}}$

RYAN

Iteration 3

Given: $(\underset{x_1}{1}, \underset{y_1}{3})$ $(\underset{x_2}{3}, \underset{y_2}{6})$

$m_2 = 1.26$, $b_2 = 1.90$

Learning rate $\alpha = 0.1$, $n = 2$

STEP 1 – Predicted values ($\hat{y} = mx + b$)

$\hat{y}_1 = 1.26(1) + 1.90 = 3.16$

$\hat{y}_2 = 1.26(3) + 1.90 = 5.68$

STEP 2 – Errors ($e_i = y_i - \hat{y}_i$)

$e_1 = 3 - 3.16 = -0.16$

$e_2 = 6 - 5.68 = 0.32$

STEP 3 – MSE

$MSE = (1/2)[e_1^2 + e_2^2]$

$= (1/2)[(-0.16)^2 + (0.32)^2]$

$= (1/2)[0.12680]$

$= 0.0640$

Decreased from 1.04 to 0.064

STEP 4 – Gradients

$\partial J/\partial m = 10m + 4b - 21$

$= 10(1.26) + 4(1.90) - 21$

$= -0.08$

$\partial J/\partial b = 4m + 2b - 9$

$= 4(1.26) + 2(1.90) - 9$

$= -0.16$

STEP 5 – Update

$m_3 = m_2 - \alpha(\partial J/\partial m)$

$= 1.26 - (0.1)(-0.80)$

$= 1.34$

$b_3 = b_2 - \alpha(\partial J/\partial m)$

$= 1.90 - (0.1)(-0.16)$

$= 1.916$

x	y	m_2	b_2	m_3	b_3	$\partial J/\partial m$	$\partial J/\partial b$	MSE
1	3	1.26	1.90	1.34	1.916	−0.80	−0.16	0.064
3	6							